Ad Astra Team Weekly Presentation

Hacking for Defense | Berkeley Security & Resilience Accelerator

Sponsor: SpaceWERX
Mentor: Max Dauber

Week 5 / Spring 2026

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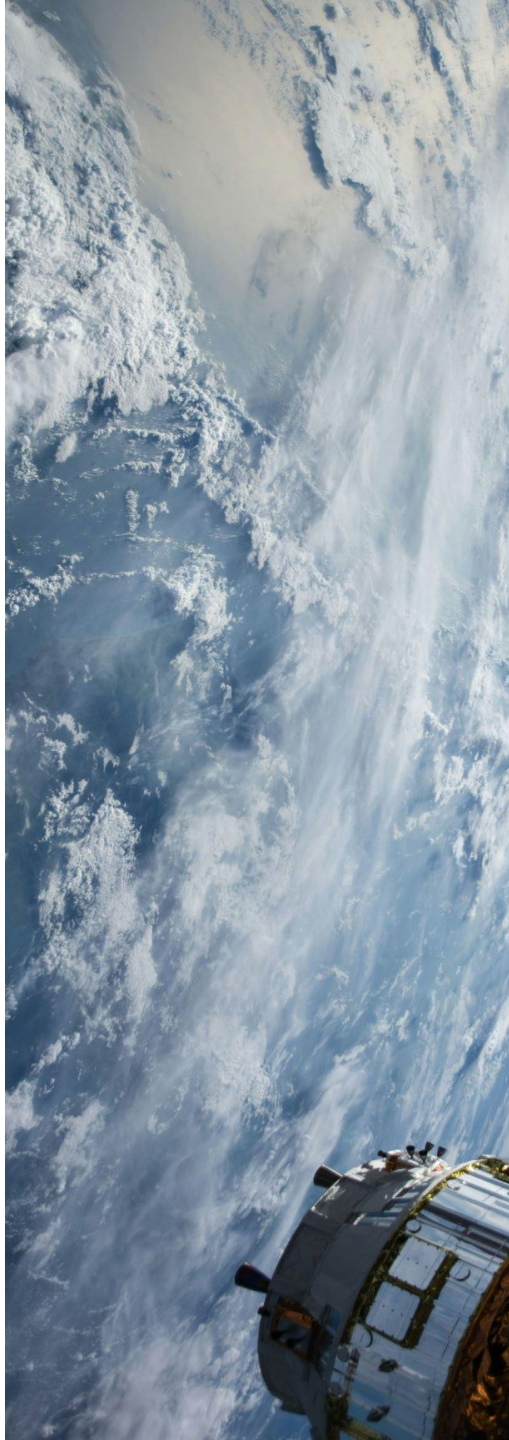


Ad Astra | Week 5

INITIAL PROBLEM STATEMENT (Provided by Sponsor)

Multiple DoW divisions are seeking interoperability solutions, **suggesting either systemic friction or policy-driven requirements.** We hypothesize that **logistics coordinators experience delays and errors when requesting and translating** between DoW's 10-class taxonomy and NASA's 6 functional categories.

UPDATED PROBLEM STATEMENT

Space Force joint programs with NASA and commercial partners experience repeated delays and rework during planning and acquisition because priorities, requirements, and risk are interpreted differently across organizations, with no formal mechanism to translate them. This translation is currently handled through informal, person-dependent workarounds rather than institutional processes, making coordination fragile and inconsistent as personnel rotate.

54
Individuals Outreached

8
Interviews Completed This Week

21
Total Interviews Completed

Beneficiary Discovery Summary

In addition to confirming last week's insights - see below-, we identified a desire for an "Interagency Working Group. Stakeholders" from both NASA and the USSF have expressed that technical tools will fail without a dedicated 'Working Table' to align operational cultures and resolve regulatory barriers early in the mission life cycle.

- **Main Friction is Weak Demand Signaling**
    - Absence of clear demand signals leads to **unclear operational prioritization**

- **Knowledge Translation is informal, relationship-driven, and non-institutionalized**
    - Interagency coordination relies on informal liaisons rather than formalized processes

- **Policy and Systems misalignment compound readiness friction**
    - Existing doctrines lack space-specific priority language in logistics systems
    - Internal logistics systems are not fully interoperable
    - Competing standards require translation layers rather than forced convergence

Mission Model Canvas

KEY PARTNERS

Jondavid Duvall - Space Logistics Officer, Space Systems Command, United States Space Force

Vincent Capello, & Taylor Macquain - Logistics Management Division Director @ Nasa

KEY ACTIVITIES

- Test the Policy Route/Working Group Route
- Vandenberg Visit
- Beneficiary Discovery with Logistics Officers
- Understand the policy "ecosystem"

KEY RESOURCES

Policy Authority: Identify a Command Policy pathway (agile authority at commander level) and Knowledge on existing legal framework NASA is adopting)

VALUE PROPOSITIONS

Establish the missing institutional bridge between Space Force and NASA through a formal Interagency Coordination Office that consolidates procurement strategy, creates unified demand signals to industry, and provides supply chain visibility

BUY-IN / SUPPORT

Jondavid Duvall - Space Logistics Officer,

Vincent Capello, & Taylor Macquain - Logistics Management Division Director @ Nasa

DEPLOYMENT

Deployment is considered successful when an Acquisition Officer can use the tool to map a NASA requirement to a DoD National Stock Number (NSN) in under 5 minutes without calling a human liaison.

BENEFICIARIES

Space Force Logistics Officer

Mission commanders requiring real-time asset visibility (USSPACECOM & USTRANSCOM)

Joint operations planners coordinating DoW-NASA missions

DoD and NASA Logistics coordinators

Supply chain officers managing re-supply ops

MISSION COST / BUDGET

- Cost N/A

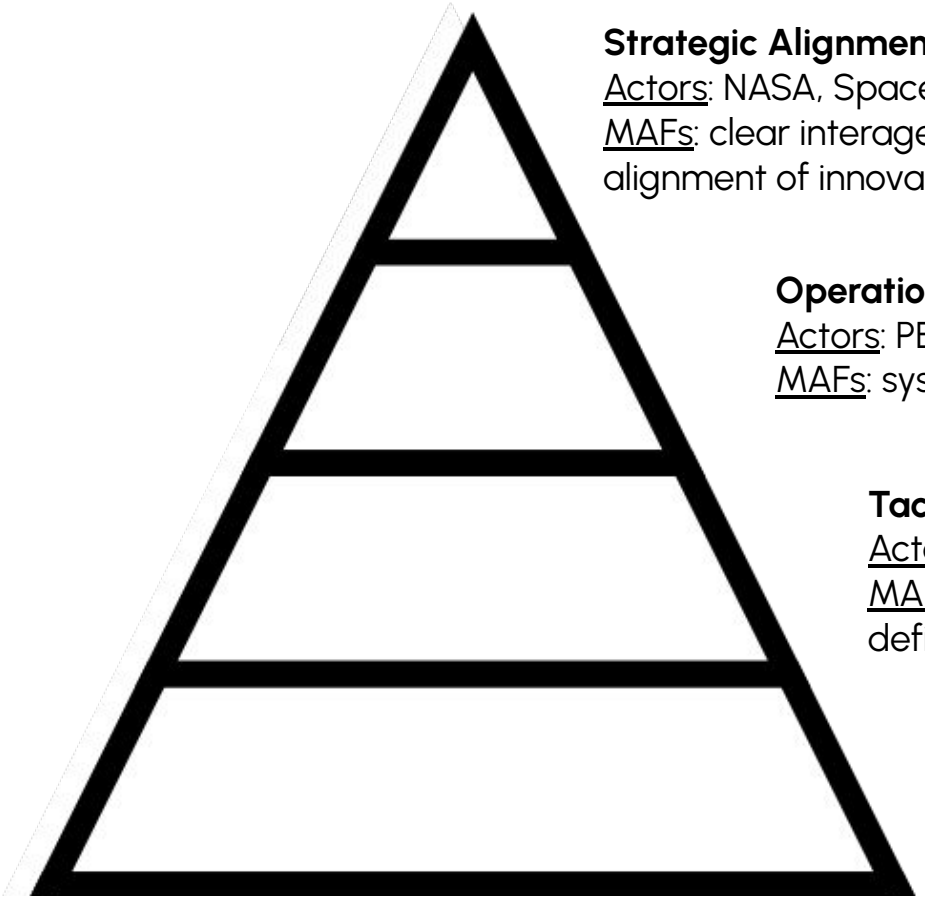
MISSION ACHIEVEMENT

Mission Success = Implementable and clear institutional coordination mechanisms between USSF and NASA that enable joint procurement strategy, supply chain visibility, and direct agency coordination

Success Metrics:

- Efficiency Gains, i.e. a reduction in planning-to-launch coordination time for joint programs
- Priority Signal clarity: Priority missions are cleared faster
- Cost savings: Reduced duplication + better commercial negotiations from unified demand

Tiered Mission Achievement



Strategic Alignment
Actors: NASA, SpaceWERX, OSD, Space Force Leadership
MAFs: clear interagency demand/priority signals, budget optimization, alignment of innovation with readiness levels

Operational Integration
Actors: PEOs, Space Operations Command (SpOC)
MAFs: system availability, integration speed, aligned readiness metrics

Tactical Execution
Actors: Operators, Launch Teams, Engineers,, Integration Teams
MAFs: clear supply criticality designation, rapid capability integration, defined sustainment ownership (contracting officer)

Sustainment & Logistics
Actors: logistics & acquisitions officers, Contractors, supply chain teams
MAFs: lifecycle traceability, interoperable classification mapping

*tiers represent decision making and not chain of command

Modified Influence Map

+Interest

HIGH INFLUENCE / HIGH INTEREST
(Key Players - Must Satisfy)

Space Systems Command Leadership (S5/S8)
→ Budget authority, sponsor, acquisition policy

US Space Command Leadership
→ Command Policy authority, operational mandate

SpaceWERX → Innovation facilitation, dual-use advocacy

NASA Logistics Management Division Director
→ Civilian agency alignment, logistic policy approval

LOW INFLUENCE / HIGH INTEREST
(Keep informed)

FFRDCs (RAND, Aerospace Corp)
→ Strategic research, policy socialization

Congressional staffers (space/defense committees)

- Influence

+ Influence

LOW INFLUENCE / LOW INTEREST
(Monitor)

International partners (NATO logistics)

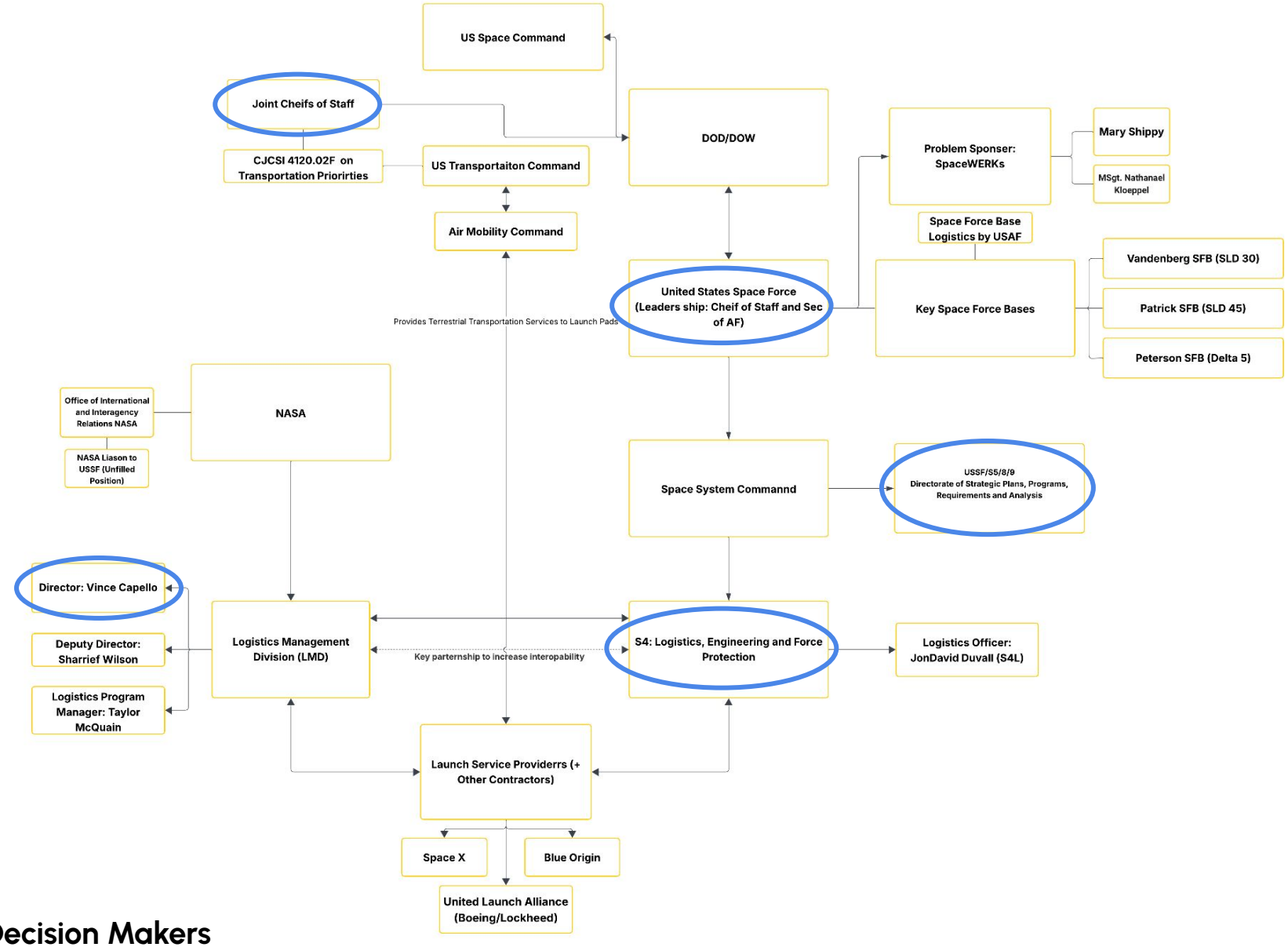
HIGH INFLUENCE / LOW INTEREST
(Keep Satisfied)

Assistant Secretary of Defense for Space Policy
→ Policy endorsement, strategic legitimacy

Joint Chiefs of Staff
→ Policy endorsement of space logistic priorities

- Interest

Graphical Influence Map



Legend: Decision Makers

Modified Beneficiary Workflow

End-User: DJ, Space Force Logistics Officer

Missions that require cross-collaboratio n between NASA and DoW

Mission supply preparation in coordination with both agency's supply team

Manual data/information translation from different supply categories from DoD to NASA and vice versa
-> Constant back and forth

Final budgeting and confirmation of mission costs.

Confirmation of correct measurements and units before placing additional orders for supplies.

Supply request sent, including supply exchange between agencies and requests with civilian vendors

Confirmation of correct units in orders upon order arrival

Legend: Decision Making Points

VPC: Space Force Logistics Officer

Value Proposition: Formalize coordination through a Command Policy-backed processes that preserve knowledge and reduce readiness-phase friction.

Customer Segment: Space Force Logistics Officer

Gain Creators
- **Translation Facilitator**
- **Facilitate Knowledge sharing and preservation**
- **Command Policy Authority**
- **Built-in traceability & auditability (critical for NASA)**

Solutions
- **Interoperability Framework/Policy**
- **Translation Engine:** NSN-to-class mapping, risk framework translator, compliance interface

Pain Relievers
- **Eliminate translation work arounds**
- **Real-Time Taxonomy Mapping:** Replaces the "Apples-to-Oranges" confusion with supply taxonomy
- **Clear rules and platforms to enact interoperability**

Gains
- Cross-agency knowledge
- Comparability and Operational Clarity
- **Formalized processes backed by Command Policy**
- **Reduced coordination friction**

Tasks:
- **Translate requirements** across incompatible DoD/NASA terminology
- **Coordinate cross-agency logistics**
- **Modernize Mission Logistics**: Adapt logistics for Cislunar, GEO, etc operations
- **Verify treaty compliance**

Pains
- **Incompatible data**
- **Unequal information between agencies** when it comes to space missions
- **Regulatory Concerns/Uncertainty**
- **Fragile translation dependent on finding/getting the right liaison.**

VPC: Mission Commanders

Value Proposition: Reduce mission planning delays through Command Policy-backed coordination processes that eliminate terminology friction and accelerate cross-agency planning cycles.

Customer Segment: Mission commanders requiring real-time asset visibility (USSPACECOM & USTRANSCOM)

Gain Creators
- Command Authority Solution: Within commander's jurisdiction for rapid implementation
- Improved Cross-Agency Clarity: Risk frameworks and requirements mapped
- Increased coordination & interoperability

Solutions
- Command Policy Guidance
- Translation Widget:

Pain Relievers
- Eliminate manual/slow translation between systems
- Real-time data feed
- Formalized Processes: Command Policy ensures consistent coordination, not ad hoc workarounds

Gains
- Command Policy-backed coordination mechanisms that work consistently
- Faster mission planning cycles with standardized coordination
- Reduced delays from cross-agency translation
- Real-time visibility of all DoD+NASA assets

Tasks:
- Execute time-sensitive ops
- Make real-time decisions on asset allocations
- Ensure mission-critical supply arrive on time

Pains
- Lack of visibility of assets available in real time
- Incompatible logistics
- Lack of common taxonomy causing miscommunication
- Delayed mission planning due to cross-agency coordination friction

VPC: Joint Operations Planners

Value Proposition: Accelerate joint mission planning through Command Policy-backed coordination that eliminates terminology translation delays.

Customer Segment: Joint operations planners coordinating DoW-NASA missions

Gain Creators
- Interoperability efficiency
- Automated asset visibility
- System integration
- Increased coordination & interoperability

Solutions
- Interoperability framework
- "Translation layer" enabling system-to-system data exchange and live updates

Pain Relievers
- Data standardization
- Automated data exchange
- Integrated, live data feedback loops
- Shared visibility

Gains
- Reduced translation time with formalized terminology mapping
- Standardized taxonomy/common framework
- Automated data exchange
- Ability to model different support options between Dow/Nasa

Tasks:
- Develop integrated op plans with both DoW and Nasa capabilities and supplies
- Coordinate logistics
- Navigate terminology and risk framework differences during planning

Pains
-Time spent translating terminology and risk classes between agencies
- Lack of visibility between both systems
- Different taxonomy for supplies (time spent translating these)

VPC: Logistics Coordinators

Value Proposition: Formalize cross-agency coordination through Command Policy + digital translation tools that eliminate manual requirement mapping, reduce onboarding time, and standardize processes across DoD/NASA

Customer Segment: Logistics coordinators managing cross-organizational supply requests.

Gain Creators
- Institutional Process
- Interoperability efficiency
- Visibility and better "request" services
- Improved search capability
- Reduced error/time-consuming verifications (while interpreting different taxonomies)

Solutions
- Command Policy Guidance
- Translation Widget
- Interagency Coordination Structure: Permanent mechanism for joint logistics coordination

Pain Relievers
- Formalized Translation Process
- Standardized Documentation: Command Policy establishes consistent frameworks
- Data and terminology standardization
- System integration
- Automated translations

Gains
- Formalized translation processes backed by Command Policy
- Automated translations
- Faster search capabilities across DoW/NASA inventories
- Reduce error from ranslations
- Reduce interoperability friction

Tasks:
- Translate requirements across DoD/NASA terminology
- Process supply requests between DoW/NASA
- Track status of joint supply ops
- Resolve discrepancies in inventories

Pains
- Coordination dependent on finding experienced liaisons or Material Control Officers
- Slow/Prone to error translation processes
- No automated status tracking
- Longer onboarding time due to different terminology learning curve

VPC: Supply Chain Officers

Value Proposition: Improve supply chain efficiency through coordination standardization with Command Policy + digital tools

Customer Segment: Supply chain officers managing resupply ops

Gain Creators
- Interoperability efficiency
- Visibility and better "request" services
- Improved search capability
- Reduced error/time-consuming verifications (while interpreting different taxonomies)

Solutions
- Interagency Coordination Structure: Improves upstream acquisition clarity for downstream planning
- Digital Translation Widget: Standardized terminology and documentation across agencies
- Cross-org.inventory platform, enabling inventory aggregation

Pain Relievers
- System integration
- Metrics that improve restock decisions, reducing cost and waste of materials

Gains
- Unified asset visibility
- Improve predictive analytics (better forecasting of what's available from the other agency)
- Improved readiness

Tasks:
- Coordinate and execute re-supply ops
- Optimize supply routes, budget and timelines
- Manage inventory

Pains
- Cannot consider an aggregate count (from the other agency) as part of their own inventory -> Risk of running out of stock?
- Incompatible taxonomy and systems

End User Beneficiary Archetype: Space Force Logistics Officer

DJ: The Innovative Logistics Officer at Space Force

Male	20+ years of experience in logistics	Tired of the internal bureaucracy and "translation" fatigue
Excited about efficiency + innovation		Wants a solution that will ignite bigger changes

- Very forward thinking: He knows that a solution today can significantly impact the coming years of space exploration.
- Has suffered the fatigue and frustration of having 2 different sets of languages and understanding,
  - He's seen a satellite not able to launch because of a simple metric conversion!
- Is open to any and every solution possible, which can reduce the clarity on what is the main problem/pain-point and who to prioritize this.
- Really wants new set of eyes to come and provide a solution, because it has almost felt as a "wicked problem*.

Mission Achievement Criteria

- Policies targeting space logistics priorities
- Reduced translation and bureaucratic rework

**A wicked problem is one that is difficult to define, has no clear solution, and perhaps not even a set of possible solutions, and the problem might be a symptom of another set of problems.. (Peters, 2017)*

End User Beneficiary Archetype: Space Force Logistics Officer

Victor: Logistics Officer at NASA

Logistician with deep institutional knowledge both in DoD and NASA

Values relationships, people and networks

Collaborative mindset (friends with DJ!)

Male

Excited about student-led innovation projects!

- Is seeking a formal interagency coordination mechanism
- Has his team already looking for solutions
- Recognizes why both agencies have worked differently and wants to adopt a "brother-sister" approach: both agencies function under the same "household" rules, but are allowed to be unique at the same time
- Can influence NASA logistics policy and documentation standards, but cannot unilaterally mandate DoD adoption (although can advocate for interagency mechanisms).

Mission Achievement Criteria

- Clear interagency priority designation
- Reduced requirement translation and rework
- Institutionalized coordination vehicles

Updated MVP: Policy Framework + Translation Widget

Note: We are focusing on testing the policy framework as a priority

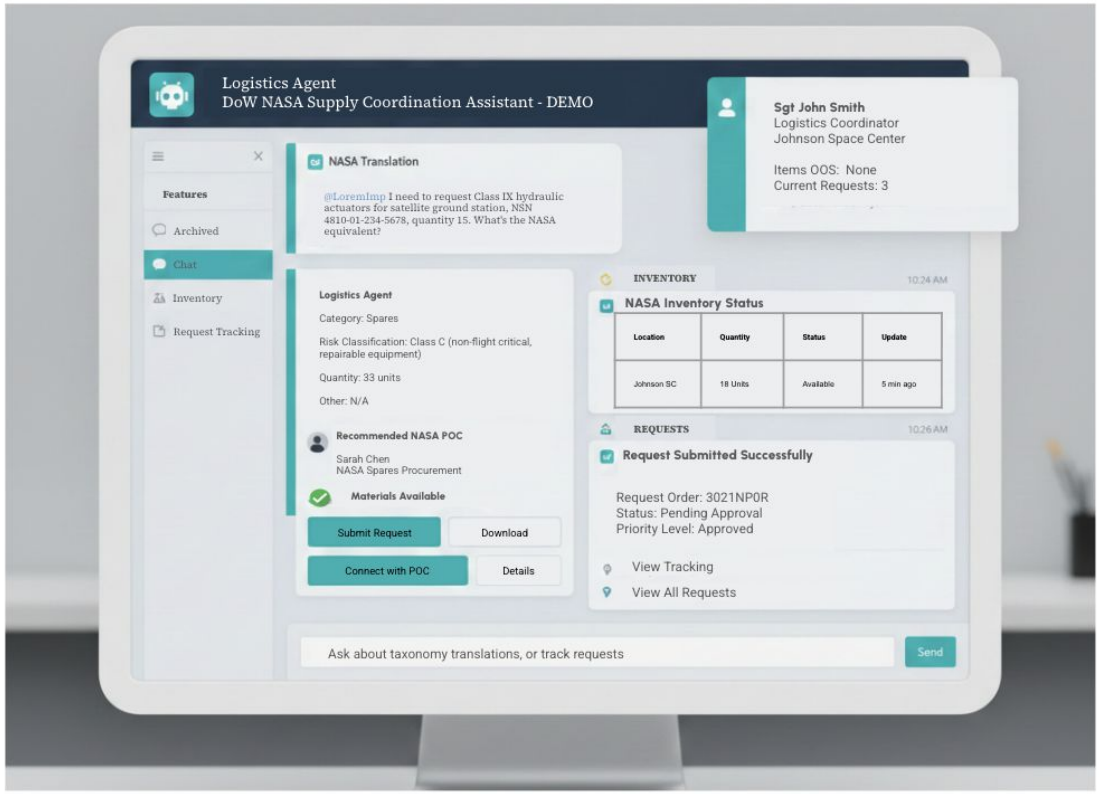
A Command Policy mandates formalized cross-agency knowledge translation processes, operationalized through a logistics translation widget that provides instant terminology/risk framework translation, shared inventory visibility, and automated request processing across Space Force-NASA operations

COMMAND POLICY FRAMEWORK (SCHEMATIC)

Purpose: Formalize cross-agency knowledge translation for Space Force-NASA operations

Key Contents

1. **Standardized Interoperability Protocol** – Mandate use of digital translation tools for all joint acquisitions
2. **Designated Translation Specialists** – Formalize Material Control Officer role with institutional backing
3. **Interagency Coordination Office** – Establish permanent Pentagon-based bridge structure
4. **Knowledge Preservation** – Capture liaison expertise in digital knowledge base, survive personnel turnover



NEXT STEPS

Beneficiary Interviews

Feb 23-Mar 2

Conversations with Space Force officers (both civilian and military), as well as NASA logistics and acquisitions officers

Mentor Meet Up

Feb 24

Work with mentor to see what angle should we "choose" based on the information, and how to extract information on what the actual problem is and the scope we should be considering

Multi-Stakeholder Workshop

Feb 23-Mar 2

Setting up and facilitating a workshop between key stakeholders to learn more about the issue from different perspectives, stakeholders' pain points, and codesign potential solutions

Testing our new MVP: Policy Workflow

Feb 23-Mar 2

This week we seek to understand how our end-users and logistic experts translate their policy recommendations into approved policy and understand what roles different decision makers play. We have meetings set up with our mentor who is assisting the Secretary of War as well as NASA's director of logistics and management.